SEPARATED MORPHISMS SATISFY THE UNIQUENESS PART OF THE VALUATIVE CRITERION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4704-4713, following the statement at 4697-4702.

Allowed standard prerequisites (editorial).

- Equalizer of two morphisms. lemma-where-are-they-equal (same chapter, cited at ~4709). The equalizer Z of a and b is a closed subscheme of Spec(A).
- Valuative criterion setup. definition-valuative-criterion (same chapter, cited at ~4705). Supplies the test diagram with A a valuation ring, K its fraction field.

Source reference. Original definition 01KD: sources/stacks-schemes/pinned-original-source.tex, source lines 3995–3995. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01KM: sources/stacks-schemes/pinned-original-source.tex, source lines 4317–4317. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. If f is separated, then f satisfies the uniqueness part of the valuative criterion.*

Proof. Let a diagram as in Definition 01KD be given. Suppose there are two morphisms $a, b : \mathrm{Spec}(A) \to X$ fitting into the diagram. Let $Z \subset \mathrm{Spec}(A)$ be the equalizer of a and b. By Lemma 01KM this is a closed subscheme of $\mathrm{Spec}(A)$. By assumption it contains the generic point of $\mathrm{Spec}(A)$. Since A is a domain this implies $Z = \mathrm{Spec}(A)$. Hence $a = b$ as desired. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.